

Firmware Update Planning Template

Document Control

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1. Firmware Update Overview

1.1 Basic Information

Item	Details
Device Type	[e.g., ASIC Miner, Network Switch, PDU, etc.]
Make and Model	[e.g., Bitmain Antminer S19 Pro]
Current Firmware Version	[e.g., v2.0.1.3]
Target Firmware Version	[e.g., v2.0.1.5]
Firmware Source	[e.g., Manufacturer website, direct from vendor, etc.]
Release Date	[YYYY-MM-DD]
Download URL	[If applicable]
Checksum/Hash	[SHA-256 hash value]
Digital Signature Verified	☐ Yes ☐ No

Number of Affected Devices	[e.g., 250 units]
Update Priority	☐ Critical ☐ High ☐ Medium ☐ Low

1.2 Firmware Release Notes Summary

[Summarize key changes, bug fixes, features, and improvements from the vendor's release notes]

2. Compatibility Analysis

2.1 Hardware Compatibility

Item	Details
Compatible Hardware Versions	[List all compatible hardware versions]
Incompatible Hardware Versions	[List if any are explicitly incompatible]
Hardware Modification Required	☐ Yes ☐ No [If yes, describe]

2.2 Software/System Compatibility

Item	Details
Compatible with Management System	☐ Yes ☐ No ☐ Unknown
Compatible with Monitoring Tools	☐ Yes ☐ No ☐ Unknown
Compatible with Mining Pools	☐ Yes ☐ No ☐ Unknown
Compatible with Other Systems	[List other system compatibility considerations]

2.3 Dependency Analysis

Dependency	Impact	Mitigation
[List any dependencies]	[Describe impact]	[Mitigation strategy]

3. Risk Assessment

3.1 Risk Analysis

Risk Category	Specific Risk	Likelihood (1-5)	Impact (1-5)	Risk Score	Mitigation Strategy
Performance	[e.g., Hashrate reduction]				
Stability	[e.g., Increased crash frequency]				
Security	[e.g., New vulnerability]				
Operational	[e.g., Pool compatibility issues]				
Financial	[e.g., Downtime cost]				
Rollback	[e.g., Rollback complexities]				

3.2 Risk Assessment Summary

[Provide an overall summary of the risk assessment and whether the update is justified based on risk vs. benefit]

4. Resource Requirements

4.1 Personnel Resources

Role	Number Required	Estimated Hours	Responsibilities
Firmware Administrator			
System Administrator			
Network Administrator			
Security Specialist			
Mining Operations			

Other			
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4.2 Technical Resources

Resource	Details	Availability
Test Environment	[Describe required test environment]	
Network Resources	[Network requirements]	
Tools Required	[List specific tools needed]	
Storage Requirements	[Any storage needs]	

4.3 Downtime Estimation

Item	Details
Estimated Downtime per Device	[e.g., 15 minutes]
Total Estimated Downtime	[e.g., 6 hours for all devices]
Mining Revenue Impact Estimate	[Estimated financial impact]

5. Testing Plan

5.1 Test Environment Setup

[Describe the test environment to be used]

5.2 Test Cases

Test ID	Test Description	Expected Result	Pass/Fail Criteria
TC-001			
TC-002			
TC-003			
[Add more as needed]			

5.3 Performance Testing

Metric	Pre-Update Baseline	Post-Update Target	Measurement Method
Hashrate			
Power Consumption			
Efficiency (J/TH)			
Temperature			
Rejection Rate			
Other Performance Metrics			

6. Deployment Strategy

6.1 Deployment Approach

Item	Details
Deployment Method	☐ Phased ☐ Canary ☐ Batch ☐ All-at-once
Deployment Sequence	[Describe the order and grouping of updates]
Automation Approach	[Detail any automation to be used]
Manual Procedures Required	[List any manual steps required]

6.2 Deployment Schedule

Phase	Devices/Groups	Date	Time	Duration	Personnel

6.3 Success Criteria

[Define clear, measurable criteria that will indicate a successful deployment]

7. Rollback Plan

7.1 Rollback Triggers

[Define specific conditions that would trigger a rollback]

7.2 Rollback Procedure

[Detail step-by-step rollback procedure]

7.3 Rollback Testing

[Describe how rollback capability will be verified before deployment]

8. Communication Plan

8.1 Pre-Deployment Communication

Stakeholder	Information to Communicate	Method	Timing	Responsible Person

8.2 During-Deployment Communication

Milestone	Stakeholders to Notify	Message	Method

8.3 Post-Deployment Communication

[Detail communication plan after deployment is complete]

9. Approval Section

9.1 Technical Approval

Role	Name	Signature	Date
Firmware Manager			

IT/Infrastructure Manager			
Security Officer			

9.2 Operational Approval

Role	Name	Signature	Date
Mining Operations Manager			
Facility Manager			
Other			

9.3 Final Authorization

Role	Name	Signature	Date
CTO/IT Director			

10. Attachment Checklist

- ☐ Vendor Release Notes
- ☐ Firmware File Hash Verification Results
- ☐ Detailed Test Plan
- ☐ Risk Assessment Documentation
- ☐ Previous Firmware Backup Verification
- ☐ Other: _____

Note: This template should be completed for all firmware updates. Sections may be adjusted based on updated criticality and scope.